

```
!pip -q install opencv-python-headless

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

print("OpenCV version:", cv2.__version__)
```

OpenCV version: 4.14.0

```
uploaded = files.upload()

image_name = next(iter(uploaded.keys()))
image = cv2.imread(image_name)

if image is None:
    raise ValueError("The uploaded file is not a valid image.")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully:", image_name)
print("Original image shape:", image.shape)
print("Grayscale image shape:", gray.shape)
```

Choose Files panda.jpg
panda.jpg(image/jpeg) - 29029 bytes, last modified: 8/26/2026 - 100% done
Saving panda.jpg to panda.jpg
Image loaded successfully: panda.jpg
Original image shape: (452, 678, 3)
Grayscale image shape: (452, 678)

```
plt.figure(figsize=(8, 6))

plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

plt.show()
```

Original Grayscale Image



```
gaussian = cv2.GaussianBlur(gray, (5, 5), 0)

plt.figure(figsize=(8, 6))

plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Blur")
plt.axis("off")

plt.show()
```

Gaussian Blur



```
median = cv2.medianBlur(gray, 5)

plt.figure(figsize=(8, 6))

plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")

plt.show()
```

Median Filter



```
average = cv2.blur(gray, (5, 5))

plt.figure(figsize=(8, 6))

plt.imshow(average, cmap="gray")
plt.title("Average (Mean) Filter")
```

```
plt.axis("off")  
  
plt.show()
```

Average (Mean) Filter



```
laplacian = cv2.Laplacian(gray, cv2.CV_64F)  
  
laplacian_abs = cv2.convertScaleAbs(laplacian)  
  
plt.figure(figsize=(8, 6))  
  
plt.imshow(laplacian_abs, cmap="gray")  
plt.title("Laplacian Filter")  
plt.axis("off")  
  
plt.show()
```

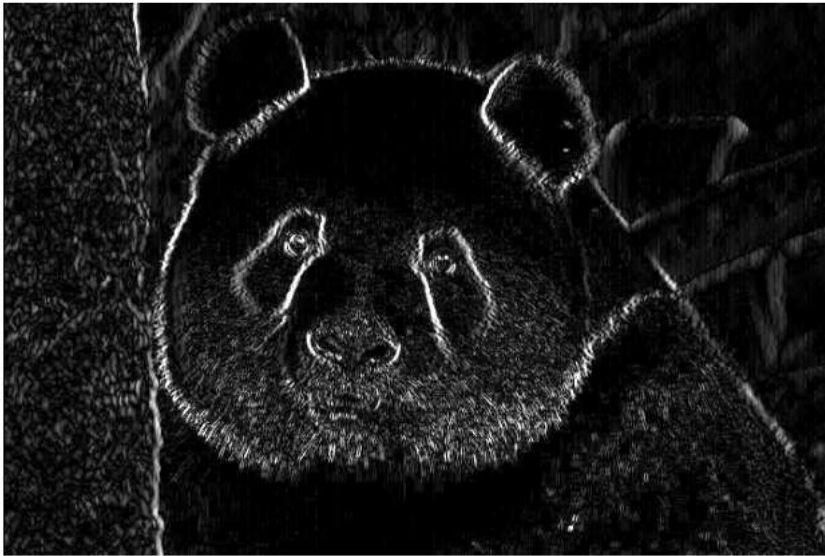
Laplacian Filter



```
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)  
  
sobel_x_abs = cv2.convertScaleAbs(sobel_x)  
  
plt.figure(figsize=(8, 6))  
  
plt.imshow(sobel_x_abs, cmap="gray")  
plt.title("Sobel X - Horizontal Edges")  
plt.axis("off")
```

```
plt.show()
```

Sobel X - Horizontal Edges



```
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_y_abs = cv2.convertScaleAbs(sobel_y)

plt.figure(figsize=(8, 6))

plt.imshow(sobel_y_abs, cmap="gray")
plt.title("Sobel Y - Vertical Edges")
plt.axis("off")

plt.show()
```

Sobel Y - Vertical Edges



```
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(laplacian_abs, cmap="gray")
plt.title("Laplacian")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(sobel_x_abs, cmap="gray")
plt.title("Sobel X")
plt.axis("off")
```

```
plt.subplot(1, 3, 3)
plt.imshow(sobel_y_abs, cmap="gray")
plt.title("Sobel Y")
plt.axis("off")

plt.tight_layout()
plt.show()
```



```
noisy = gray.copy()

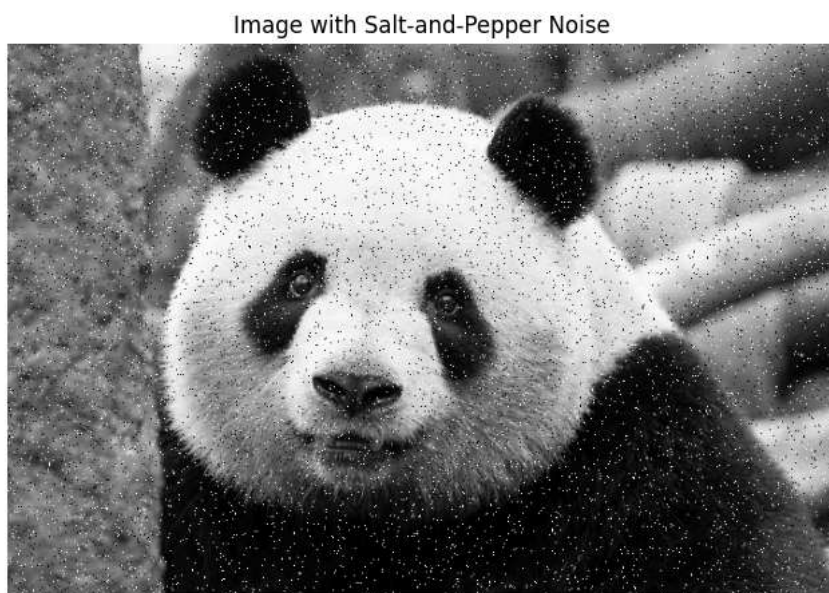
# Add salt-and-pepper noise
noise = np.random.rand(*gray.shape)

noisy[noise < 0.02] = 0
noisy[noise > 0.98] = 255

plt.figure(figsize=(8, 6))

plt.imshow(noisy, cmap="gray")
plt.title("Image with Salt-and-Pepper Noise")
plt.axis("off")

plt.show()
```



```
median_noisy = cv2.medianBlur(noisy, 5)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(noisy, cmap="gray")
plt.title("Noisy Image")
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
plt.imshow(median_noisy, cmap="gray")
plt.title("After Median Filtering")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Noisy Image



After Median Filtering



```
plt.figure(figsize=(15, 10))

plt.subplot(2, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Blur")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(average, cmap="gray")
plt.title("Average Filter")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(laplacian_abs, cmap="gray")
plt.title("Laplacian")
plt.axis("off")

plt.subplot(2, 3, 6)
plt.imshow(sobel_x_abs, cmap="gray")
plt.title("Sobel X")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original



Gaussian Blur



Median Filter



Average Filter



Laplacian



Sobel X



```
plt.figure(figsize=(8, 6))

plt.imshow(sobel_y_abs, cmap="gray")
plt.title("Sobel Y - Vertical Edges")
plt.axis("off")

plt.show()
```

Sobel Y - Vertical Edges

